

Skill-Biased Technological Change

GLOSSARY 05 - SHEET 3 / 8

Technical change that raises demand for some skills and lowers it for others

PROFESSIONAL DEFINITION

Skill-biased technological change is technical change that shifts relative demand across skill groups, raising the productivity and therefore the wages of favoured groups relative to others. The refined task-based account holds that the bias operates on tasks rather than on skill levels: technologies substitute for routine tasks - whether manual or cognitive - and complement non-routine tasks, which produces polarisation rather than a simple skill gradient (Autor, Levy and Murnane, 2003; Acemoglu and Autor, 2011).

WHO DEFINED IT ORIGINALLY

The canonical skill-biased account is Katz and Murphy's (1992). David H. Autor, Frank Levy and Richard J. Murnane reformulated it as the routine-biased task model in the Quarterly Journal of Economics in 2003; Daron Acemoglu and Autor (2011) developed the general task framework, and Acemoglu and Restrepo (2019) added the displacement-reinstatement distinction.

WHY THIS DEFINITION WAS DEVELOPED

The simple skill-based account predicted monotonically rising returns to education, but observed wage distributions were polarising: employment and wages grew at both ends and hollowed in the middle. Relocating the bias from skill to task explained this, because routine tasks are concentrated in the middle of the wage distribution rather than at its bottom.

FIGURE 1 - SUBSTITUTION AND COMPLEMENTARITY BY TASK

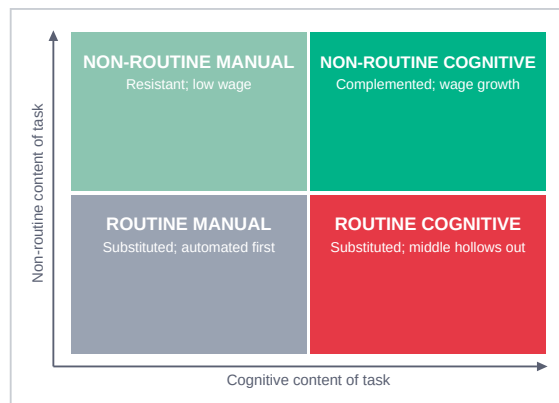


Figure 1. Substitution follows routineness, not skill level. Because routine cognitive tasks sit in the middle of the wage distribution, the effect is polarisation - and generative systems have extended substitution into parts of the upper-right quadrant previously assumed safe (after Autor, Levy and Murnane, 2003).

HOW IT IS USED

Decompose roles into tasks and classify each as routine or non-routine before estimating exposure; the task composition of a role, not its wage level or educational requirement, predicts which parts of it a system will substitute.

WHEN IT IS USED

In workforce planning ahead of AI deployment, in labour-market and policy analysis, in reskilling design, and in public debate about which occupations are exposed.

BY WHOM IT IS USED

Labour economists and policy analysts, chief human resources officers planning workforce transition, works councils, and educators designing curricula.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor model treated skill as a single ordered dimension and could not accommodate polarisation without ad hoc additions. The task framework replaces skill with task as the unit that technology substitutes or complements, which generates polarisation as a prediction rather than an anomaly. Acemoglu and Restrepo's later distinction between displacement and reinstatement effects adds the crucial policy point: whether technology reduces labour demand depends on whether new tasks are created, which is a design and investment choice rather than a technological inevitability.

RELATED DEFINITIONS IN THIS COURSE

Labour Market Displacement - Unit 5

Reskilling and Upskilling - Unit 4

Productivity Paradox - Unit 5

Digital Divide - Unit 5

General Purpose Technology - Unit 5

SOURCES (HARVARD REFERENCING STYLE)

- Acemoglu, D. and Autor, D. (2011) 'Skills, tasks and technologies: implications for employment and earnings', in Ashenfelter, O. and Card, D. (eds) Handbook of Labor Economics, Vol. 4B. Amsterdam: Elsevier, pp. 1043-1171.
- Acemoglu, D. and Restrepo, P. (2019) 'Automation and new tasks: how technology displaces and reinstates labor', Journal of Economic Perspectives, 33(2), pp. 3-30.
- Autor, D.H., Levy, F. and Murnane, R.J. (2003) 'The skill content of recent technological change: an empirical exploration', Quarterly Journal of Economics, 118(4), pp. 1279-1333.
- Katz, L.F. and Murphy, K.M. (1992) 'Changes in relative wages, 1963-1987: supply and demand factors', Quarterly Journal of Economics, 107(1), pp. 35-78.